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AUTOMATIC WATER REFILLER
For Air-Coolers

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Air-coolers provide cool air in a room by adding moisture to the air. This moist air is blown inside the room by an exhaust fan or blower, which makes the room temperature drop. These coolers require frequent refilling of water. Presented here is a circuit to refill water automatically in an air-cooler tank when the water drops below a predetermined level.

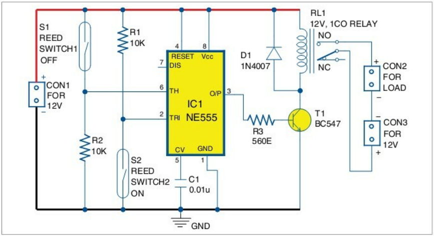

Fig. 1: Circuit diagram of automatic water refiller for air-coolers

Circuit and working

Circuit diagram of the automatic water refiller for air-coolers is shown in Fig. 1. It is built around NE555 timer (IC1), transistor BC547 (T1), reed switches (S1 and S2), and a few other components. IC1 is configured similar to bi-stable mode, the only difference being that switch S1 is connected to pin 6 instead of pin 4.

The two reed switches (S1 and S2) are affixed at upper and bottom levels of the tank as shown in Fig. 2. Working of the circuit is pretty simple.

If water drops below the predetermined level, the magnetic float closes reed switch S2 to pull pin 2 of IC1 to ground. Thereby the voltage on pin 2 goes below $1/3V_{cc}$ and the output of IC1 goes high. This energises the relay to activate the solenoid valve. Thus the water flows into the tank.

When the tank is full, the magnetic float closes reed switch S1. Pin 6 of IC1 goes above $2/3V_{cc}$ and the output goes low. This de-energises the relay to deactivate the solenoid valve, which stops the water flowing into the cooler tank.

Construction and testing

An actual-size PCB layout of the automatic water refiller for air-coolers is shown in Fig. 3 and its components layout in Fig. 4. Assemble the circuit and enclose it in a suitable cabinet.

The water tank of an air-cooler has two holes—drain outlet and overflow outlet. Level-sensing arrangement for the water tank is done as follows: Attach a small-diameter PVC pipe to the tank (refer Fig. 2) between the two outlets. Put a magnetic float inside the PVC pipe. Affix one reed switch on the pipe near the drain outlet and the other reed switch near the overflow outlet.

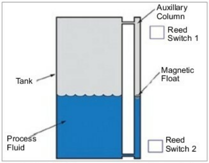

Fig. 2: Reed switch arrangement on the cooler tank

The magnetic float can be made